WORKED SOLUTION

STEP II 2004, Question 3

Pure • Curve Sketching

STEP II 2004, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s2-q3

The key idea

Everything comes from one curve. For C, factorise the derivative completely: the repeated factor $(x-2)$ survives differentiation, and what is left is a quadratic. Then the other two curves are built from C rather than started from scratch: in (i) we take $y = \pm\sqrt{f(x)}$, and in (ii) we feed x^2 into f. Keep track of where f is positive, where it has simple roots and where it has a repeated root; those details decide the shapes.

Solution

Write $f(x) = x(x+1)(x-2)^4$, so C is $y = f(x)$.

Stationary points of C

By the product rule, taking out the common factor $(x-2)^3$,

$$\begin{aligned} f'(x) &= (2x+1)(x-2)^4 + 4x(x+1)(x-2)^3 \\ &= (x-2)^3[(2x+1)(x-2) + 4x^2 + 4x] \\ &= (x-2)^3(6x^2 + x - 2) \\ &= (x-2)^3(3x+2)(2x-1). \end{aligned}$$

So the stationary points are at $x = -\frac{2}{3}$, $x = \frac{1}{2}$ and $x = 2$, with

$$f(-\tfrac{2}{3}) = (-\tfrac{2}{3})(\tfrac{1}{3})(-\tfrac{8}{3})^4 = -\frac{8192}{729}, \qquad f(\tfrac{1}{2}) = \tfrac{1}{2}\cdot\tfrac{3}{2}\cdot(-\tfrac{3}{2})^4 = \frac{243}{64}, \qquad f(2) = 0.$$

To classify them, look at the sign of each factor of f':

	$x < -\frac{2}{3}$	$-\frac{2}{3} < x < \frac{1}{2}$	$\frac{1}{2} < x < 2$	$x > 2$
$(x-2)^3$	$-$	$-$	$-$	$+$
$(3x+2)(2x-1)$	$+$	$-$	$+$	$+$
$f'(x)$	$-$	$+$	$-$	$+$

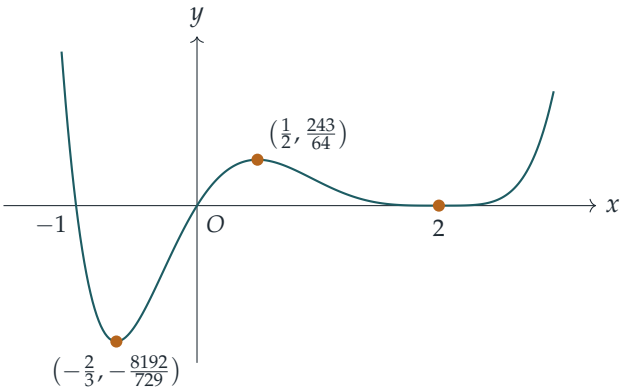
Hence

- $\left(-\frac{2}{3}, -\frac{8192}{729}\right)$ is a minimum (f' goes from $-$ to $+$);
- $\left(\frac{1}{2}, \frac{243}{64}\right)$ is a maximum (f' goes from $+$ to $-$);
- $(2, 0)$ is a minimum (f' goes from $-$ to $+$).

The last one is also clear without calculus: $f(x) \geq 0$ near $x = 2$ because $x(x+1) > 0$ there and $(x-2)^4 \geq 0$.

Sketching C

The curve crosses the x-axis at $x = -1$ and $x = 0$ (simple roots, with gradients $f'(-1) = -81$ and $f'(0) = 16$) and touches it at $x = 2$. Near $x = 2$, $f(x) \approx 2 \cdot 3\,(x-2)^4 = 6(x-2)^4$, so the curve is very flat there. Since f is a polynomial of degree 6 with positive leading coefficient, $y \to +\infty$ as $x \to \pm\infty$. The vertical scale in the sketch is compressed.



Part (i)

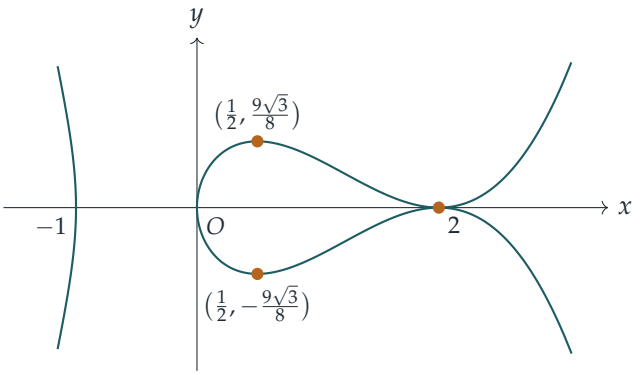
Now $y^2 = f(x)$, so $y = \pm\sqrt{f(x)}$. Three facts settle the shape.

- *Where the curve exists.* We need $f(x) \geq 0$, which from C means $x \leq -1$ or $x \geq 0$. There is no curve for $-1 < x < 0$.
- *Symmetry.* Replacing y by $-y$ leaves the equation unchanged, so the curve is symmetric in the x-axis. It is enough to sketch the upper half $y = \sqrt{f(x)} = (x-2)^2\sqrt{x(x+1)}$ and reflect it.
- *Stationary points.* For $y = \sqrt{f}$ with $f > 0$ we have $\mathrm{d}y/\mathrm{d}x = f'/(2\sqrt{f})$, so the upper half is stationary exactly where $f' = 0$ and $f > 0$. That leaves only $x = \frac{1}{2}$ (the point $x = -\frac{2}{3}$ is outside the domain), giving a maximum at $\left(\frac{1}{2}, \sqrt{243/64}\right) = \left(\frac{1}{2}, \frac{9\sqrt{3}}{8}\right)$, and a minimum at $\left(\frac{1}{2}, -\frac{9\sqrt{3}}{8}\right)$ on the lower half.

Ask yourself What happens at $(-1,0)$, $(0,0)$ and $(2,0)$, where $f = 0$?

At a *simple* root such as $x = 0$, $f(x) \approx 16x$, so near the origin $y^2 \approx 16x$: a sideways parabola, and the curve crosses the axis with a *vertical* tangent. The same happens at $x = -1$, where $y^2 \approx 81(-1-x)$. At the *repeated* root $x = 2$, $y^2 \approx 6(x-2)^4$, so $y \approx \pm\sqrt{6}\,(x-2)^2$: two parabola-like arcs, one above and one below the axis, both touching the x-axis at $(2,0)$. So the upper and lower halves meet there with a common horizontal tangent; the upper half has a minimum and the lower half a maximum at $(2,0)$.

For large $|x|$, $y \approx \pm|x|^3$. So the curve is a closed loop from $(0,0)$ to $(2,0)$, pinched at $(2,0)$, continuing into two branches that open out to the right, together with a separate piece through $(-1,0)$ that opens out to the left.



Part (ii)

Here $y = x^2(x^2+1)(x^2-2)^4 = f(x^2)$. Two observations do most of the work.

- The curve is symmetric in the y-axis, since x appears only as x^2.
- As x runs over $[0,\infty)$, $u = x^2$ runs over $[0,\infty)$ increasing, so for $x \geq 0$ the curve is the part of C with $x \geq 0$, re-parametrised. In particular $y \geq 0$ everywhere, and the stretch of C with negative x (including its minimum) plays no part.

By the chain rule,

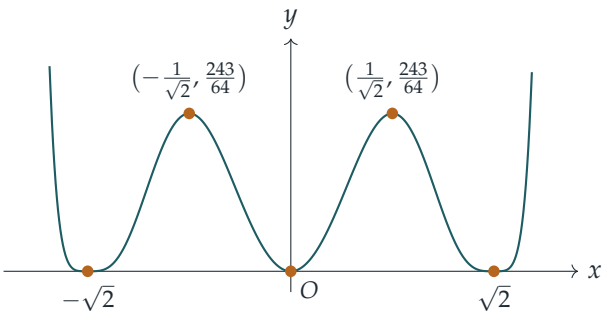
$$\frac{\mathrm{d}y}{\mathrm{d}x} = 2x\,f'(x^2) = 2x\,(x^2-2)^3(3x^2+2)(2x^2-1),$$

which vanishes when $x = 0$, $x^2 = \frac{1}{2}$ or $x^2 = 2$ (the factor $3x^2+2$ is never zero). On $u \geq 0$, f rises from 0 to its maximum at $u = \frac{1}{2}$, falls to 0 at $u = 2$ and then rises for ever, and $u = x^2$ preserves this order for $x \geq 0$. So the stationary points are

- a minimum at $(0,0)$, where $y \approx 16x^2$;
- maxima at $\left(\pm\frac{1}{\sqrt{2}}, \frac{243}{64}\right)$;

- minima at $(\pm\sqrt{2},\, 0)$, where the curve touches the axis very flatly: $y \approx 384\,(x \mp \sqrt{2})^4$.

For large $|x|$, $y \approx x^{12}$, which rises very steeply.



Checks

A quick sanity check on the derivative: at $x = 0$, $f(x) = x(x+1)(x-2)^4 \approx 16x$, and indeed $f'(0) = (-8)(2)(-1) = 16$. A numerical check confirms the factorisation of f' against a finite-difference derivative at random points, gives $f(-\frac{2}{3}) \approx -11.237$ and $f(\frac{1}{2}) = 3.796875$, and finds the largest value of $\sqrt{f}$ on $[0,2]$ to be $1.94856\ldots = \frac{9\sqrt{3}}{8}$, at $x = 0.5$. For (ii), $\mathrm{d}y/\mathrm{d}x$ changes sign at $x \approx \pm 0.707$ and ± 1.414 (and at 0), as predicted.

Common slips

- Expanding f before differentiating. Then f' is a quintic with no visible factors; keeping $(x-2)^3$ outside leaves only a quadratic to solve.
- In (i), drawing a cusp or a corner at $(0,0)$ and $(-1,0)$. At a simple root of f the tangent to $y^2 = f(x)$ is vertical.
- In (ii), copying the minimum of C at $x = -\frac{2}{3}$. Since $x^2 \geq 0$, only the part of C with $x \geq 0$ is used, and y is never negative.

Beyond STEP

The behaviour of $y^2 = f(x)$ near a root of f is the beginning of the study of singular points of algebraic curves: a simple root gives a smooth point with a vertical tangent, while a repeated root gives a place where two branches meet, as at $(2,0)$. When f is a cubic, curves $y^2 = f(x)$ are *elliptic curves*, central in modern number theory. For a careful treatment of the calculus underlying curve sketching (monotonicity, the sign of the derivative and the chain rule), see Abbott, *Understanding Analysis*.

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